

Newton's Method and Modifications

Chapter 10 of Course Notes

Lecture 12

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Course Flow: Gradient-Based Unconstrained Optimization

- Lecture 11: Steepest Descent Method
 - Simple first-order method (uses only gradients)
 - Linear convergence rate (slow on ill-conditioned problems)

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 - Second-order method (uses curvature via Hessian)
 - Quadratic convergence rate (extremely fast near solution)
 - Modifications for robustness: line search, Levenberg-Marquardt

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- Lecture 11: Steepest Descent Method
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- Lecture 12 (Today): Newton's Method and Modifications
 - Second-order method (uses curvature via Hessian)
 - Quadratic convergence rate (extremely fast near solution)
 - Modifications for robustness: line search, Levenberg-Marquardt
- Lectures 13-15: Quasi-Newton and Conjugate Gradient Methods
 - Approximate Hessian from gradient information
 - Superlinear convergence without explicit Hessian computation

Conjugate Directions

- Definition of Conjugate Directions

- Linear Independence of Conjugate Directions

- How to Generate Conjugate Directions

Conjugate Direction Method for Quadratic Programs

- The Problem and Algorithm

- Exact Step Size in the Conjugate Direction Method

- Key Property of the Conjugate Direction Method

- Numerical Example

- Exercise and Solution

The Conjugate Gradient Method

- Motivation and Direction Formula

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Why Newton's Method?

Limitation of Steepest Descent

- Uses only first-order information ($\nabla f(\mathbf{x}_k)$)
- Misses important information about curvature
- Convergence rate is linear: $\|\mathbf{x}_{k+1} - \mathbf{x}^*\| \leq \rho \|\mathbf{x}_k - \mathbf{x}^*\|$
- For ill-conditioned problems: very slow convergence

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Key Idea: Local Quadratic Approximation

At each iteration, minimize the **second-order Taylor expansion** of f :

$$m_k(\mathbf{d}) = f(\mathbf{x}_k) + \nabla f(\mathbf{x}_k)^T \mathbf{d} + \frac{1}{2} \mathbf{d}^T \nabla^2 f(\mathbf{x}_k) \mathbf{d}$$

The minimizer is:

$$\mathbf{d}_k^N = -[\nabla^2 f(\mathbf{x}_k)]^{-1} \nabla f(\mathbf{x}_k)$$

This gives **quadratic convergence** near the solution!

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Newton's Method Algorithm

Algorithm: Newton's Method

Initialization: Choose starting point $\mathbf{x}_1 \in \mathbb{R}^n$ and tolerance $\epsilon > 0$.

1. If $\|\nabla f(\mathbf{x}_k)\|_2 < \epsilon$, **STOP**.
2. Compute Newton direction:

$$\mathbf{d}_k = -[\nabla^2 f(\mathbf{x}_k)]^{-1} \nabla f(\mathbf{x}_k)$$

3. Line search (optional): $\lambda_k = \text{backtracking}(\mathbf{x}_k, \mathbf{d}_k)$, or set $\lambda_k = 1$.
4. Update: $\mathbf{x}_{k+1} = \mathbf{x}_k + \lambda_k \mathbf{d}_k$. Set $k \leftarrow k + 1$, go to Step 1.

Computational cost per iteration:

- Compute Hessian: $O(n^2)$ or $O(n^3)$ (problem dependent)
- Solve linear system: $O(n^3)$ (LU factorization)
- Total: **Much more expensive** than steepest descent

Newton Direction is a Descent Direction

Theorem 2.1: Theorem

If $\nabla^2 f(\mathbf{x}_k) \succ 0$ (positive definite) and $\nabla f(\mathbf{x}_k) \neq \mathbf{0}$, then the Newton direction

$$\mathbf{d}_k^N = -[\nabla^2 f(\mathbf{x}_k)]^{-1} \nabla f(\mathbf{x}_k)$$

is a descent direction.

Proof:

1. The directional derivative is:

$$\nabla f(\mathbf{x}_k)^T \mathbf{d}_k^N = -\nabla f(\mathbf{x}_k)^T [\nabla^2 f(\mathbf{x}_k)]^{-1} \nabla f(\mathbf{x}_k)$$

2. Let $H := \nabla^2 f(\mathbf{x}_k)$ and $H := \nabla^2 f(\mathbf{x}_k)$. Since $H \succ 0$, we have $H^{-1} \succ 0$.
3. Therefore: $H^{-1} \succ 0$, so

$$\nabla f(\mathbf{x}_k)^T \mathbf{d}_k^N = -\nabla f(\mathbf{x}_k)^T H^{-1} \nabla f(\mathbf{x}_k) < 0$$

This confirms descent. $\square$

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Newton's Method on Quadratic Functions

For convex quadratic: $f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T Q \mathbf{x} + \mathbf{c}^T \mathbf{x}$ with $Q \succ 0$

- Gradient: $\nabla f(\mathbf{x}) = Q\mathbf{x} + \mathbf{c}$
- Hessian: $\nabla^2 f(\mathbf{x}) = Q$ (constant)
- Newton direction: $\mathbf{d}_0^N = -Q^{-1}(Q\mathbf{x}_0 + \mathbf{c})$
- Newton step: $\mathbf{x}_1 = \mathbf{x}_0 + \mathbf{d}_0^N = -Q^{-1}\mathbf{c} = \mathbf{x}^*$

Result: Newton reaches the global optimum in **exactly ONE step**, regardless of starting point!

Comparison with Steepest Descent:

- Newton: 1 iteration (**unbeatable!**)
- **Steepest Descent:** $O(\kappa(Q) \log(1/\epsilon))$ iterations

Why? Newton exploits the **quadratic structure** via the Hessian; steepest

Numerical Example: Newton vs. Steepest Descent

Problem: $\min_{(x_1, x_2)} (x_1 - 2)^4 + (x_1 - 2x_2)^2$

Convergence Comparison:

Iteration	Newton $\ \nabla f(\mathbf{x}_k)\ $	Steepest Descent $\ \nabla f(\mathbf{x}_k)\ $	Ratio
0	8.944	8.944	1.0
1	1.073	3.162	0.34
2	0.013	1.414	0.009
3	2×10^{-6}	0.707	3×10^{-6}
4	$< 10^{-15}$	0.383	—
Converged?	Yes (4 iters)	No (40+ needed)	

Key observation: Newton shows **quadratic convergence**: error squares at each iteration once close to solution. Steepest descent improves linearly.

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Convergence Rate Theorem

Theorem 4.1: Theorem: Quadratic Convergence of Newton's Method

Let f be twice continuously differentiable with $\mathbf{x}^*$ a local minimum satisfying $\nabla f(\mathbf{x}^*) = \mathbf{0}$ and $\nabla^2 f(\mathbf{x}^*) \succ 0$.

There exists a neighborhood $\mathcal{N}$ of $\mathbf{x}^*$ such that if $\mathbf{x}_0 \in \mathcal{N}$, Newton's method (without line search) converges to $\mathbf{x}^*$ with **quadratic convergence**:

$$\|\mathbf{x}_{k+1} - \mathbf{x}^*\| \leq C \|\mathbf{x}_k - \mathbf{x}^*\|^2$$

for some constant $C > 0$ depending on $\nabla^2 f(\mathbf{x}^*)$ and the Hessian's Lipschitz constant.

Interpretation: Once close enough to $\mathbf{x}^*$, the **number of correct digits doubles** at each iteration!

- If $\|\mathbf{x}_0 - \mathbf{x}^*\| = 10^{-1}$ and $C = 1$:
 - $\|\mathbf{x}_1 - \mathbf{x}^*\| \leq 10^{-2}$
 - $\|\mathbf{x}_2 - \mathbf{x}^*\| \leq 10^{-4}$
 - $\|\mathbf{x}_3 - \mathbf{x}^*\| \leq 10^{-8}$

Proof: Quadratic Convergence (Part 1)

Assumptions:

- $\nabla f(\mathbf{x}^*) = \mathbf{0}$ and $H^* := \nabla^2 f(\mathbf{x}^*) \succ 0$
- The Hessian is Lipschitz continuous:
$$\|\nabla^2 f(\mathbf{x}) - \nabla^2 f(\mathbf{y})\| \leq L\|\mathbf{x} - \mathbf{y}\|$$

Step 1: Error Recursion

Let $\mathbf{e}_k := \mathbf{x}_k - \mathbf{x}^*$. The Newton update gives:

$$\begin{aligned}\mathbf{e}_{k+1} &= \mathbf{x}_{k+1} - \mathbf{x}^* = \mathbf{x}_k - [\nabla^2 f(\mathbf{x}_k)]^{-1} \nabla f(\mathbf{x}_k) - \mathbf{x}^* \\ &= \mathbf{e}_k - [\nabla^2 f(\mathbf{x}_k)]^{-1} \nabla f(\mathbf{x}_k)\end{aligned}$$

Step 2: Taylor Expansion of Gradient

By integral form (since $\nabla f(\mathbf{x}^*) = \mathbf{0}$):

$$\nabla f(\mathbf{x}_k) = \left(\int_0^1 \nabla^2 f(\mathbf{x}^* + t \mathbf{e}_k) dt \right) \mathbf{e}_k =: \bar{H}_k \mathbf{e}_k$$

Proof: Quadratic Convergence (Part 2)

Step 3: Substitute into Error Recursion

$$\mathbf{e}_{k+1} = \mathbf{e}_k - [\nabla^2 f(\mathbf{x}_k)]^{-1} \tilde{H}_k \mathbf{e}_k = [\nabla^2 f(\mathbf{x}_k)]^{-1} [\nabla^2 f(\mathbf{x}_k) - \tilde{H}_k] \mathbf{e}_k$$

Step 4: Bound Using Lipschitz Continuity

$$\begin{aligned} \|\nabla^2 f(\mathbf{x}_k) - \tilde{H}_k\| &= \left\| \int_0^1 [\nabla^2 f(\mathbf{x}^* + \mathbf{e}_k) - \nabla^2 f(\mathbf{x}^* + t \mathbf{e}_k)] dt \right\| \\ &\leq \int_0^1 L \|(1-t)\mathbf{e}_k\| dt = L \|\mathbf{e}_k\| \int_0^1 (1-t) dt \\ &= \frac{L}{2} \|\mathbf{e}_k\| \end{aligned}$$

Step 5: Take Norms

$$\|\mathbf{e}_{k+1}\| \leq \|[\nabla^2 f(\mathbf{x}_k)]^{-1}\| \cdot \frac{L}{2} \|\mathbf{e}_k\|^2$$

Near $\mathbf{x}^*$: $\|[\nabla^2 f(\mathbf{x}_k)]^{-1}\| \rightarrow \|[H^*]^{-1}\|$, so

$$\boxed{\|\mathbf{e}_{k+1}\| \leq \underbrace{\frac{L}{2} \|[H^*]^{-1}\|}_{=: C} \|\mathbf{e}_k\|^2}$$

This proves quadratic convergence provided $C \|\mathbf{e}_0\| < 1$. $\square$

Key Insight: Local vs. Global Convergence

The "Local" Convergence Caveat

The quadratic convergence theorem requires $\mathbf{x}_0$ sufficiently close to $\mathbf{x}^*$.

- Starting far away: Newton may diverge or make slow progress
- Starting close: Newton exhibits superfast quadratic convergence

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Solution: Use Line Search for Global Convergence

Combine Newton's direction with a backtracking or Wolfe line search:

1. Compute Newton direction: $\mathbf{d}_k = -[\nabla^2 f(\mathbf{x}_k)]^{-1} \nabla f(\mathbf{x}_k)$
2. Line search: Find $\lambda_k > 0$ ensuring sufficient decrease
3. Update: $\mathbf{x}_{k+1} = \mathbf{x}_k + \lambda_k \mathbf{d}_k$

Result:

- Global convergence: works from any starting point
- Local quadratic: once close to $\mathbf{x}^*$, $\lambda_k \rightarrow 1$ and quadratic rate

Advantages and Disadvantages of Newton's Method

Advantages:

- Quadratic convergence (extremely fast when close)
- One step on quadratic problems
- Scale-invariant (unlike steepest descent)
- Handles ill-conditioned problems naturally
- Easy interpretation via Taylor expansion

Disadvantages:

- **High cost:** $O(n^3)$ per iteration (system solve)
- **Requires Hessian:** expensive or unavailable
- **Hessian may not be PD:** direction fails
- **Local convergence only:** needs good starting point (or line search)
- **Storage:** $O(n^2)$ for Hessian matrix

When to Use Newton's Method:

- Small to medium problems ($n \leq 10^4$)

Exercise 1: One Step of Newton's Method

One Step of Newton's Method For $f(x_1, x_2) = x_1^2 + 4x_2^2$ with $\mathbf{x}_0 = (4, 2)^T$:

1. Compute $\nabla f(\mathbf{x}_0)$ and $\nabla^2 f(\mathbf{x}_0)$
2. Compute the Newton direction $\mathbf{d}_0^N = -[\nabla^2 f(\mathbf{x}_0)]^{-1} \nabla f(\mathbf{x}_0)$
3. Compute $\mathbf{x}_1 = \mathbf{x}_0 + \mathbf{d}_0^N$
4. Compare with steepest descent: which reaches $\mathbf{x}^* = (0, 0)$ faster?

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Solution $f(x_1, x_2) = x_1^2 + 4x_2^2$, optimal point: $\mathbf{x}^* = (0, 0)$

(1) $\nabla f(\mathbf{x}) = (2x_1, 8x_2)^T$, so $\nabla f(\mathbf{x}_0) = (8, 16)^T$

$$\nabla^2 f(\mathbf{x}) = \begin{pmatrix} 2 & 0 \\ 0 & 8 \end{pmatrix} \text{ (constant!)}$$

$$(2) [\nabla^2 f(\mathbf{x}_0)]^{-1} = \begin{pmatrix} 1/2 & 0 \\ 0 & 1/8 \end{pmatrix}, \text{ so } \mathbf{d}_0^N = - \begin{pmatrix} 4 \\ 2 \end{pmatrix}$$

$$(3) \mathbf{x}_1 = (4, 2)^T + (-4, -2)^T = (0, 0)^T = \mathbf{x}^* \checkmark$$

(4) Newton: 1 step. Steepest descent: many steps. For quadratics, Newton is unbeatable!

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Newton's Method with Line Search

Problem: Without line search, Newton can diverge far from the solution.

Solution: Combine Newton direction with line search

Algorithm: Newton with Line Search

1. If $\|\nabla f(\mathbf{x}_k)\| < \epsilon$, **STOP**.
2. Compute Newton direction: $\mathbf{d}_k = -[\nabla^2 f(\mathbf{x}_k)]^{-1} \nabla f(\mathbf{x}_k)$
3. **Line search:** Find λ_k (e.g., via backtracking or Wolfe conditions)
4. Update: $\mathbf{x}_{k+1} = \mathbf{x}_k + \lambda_k \mathbf{d}_k$. Set $k \leftarrow k + 1$, repeat.

Key Properties:

- If $\nabla^2 f(\mathbf{x}_k) \succ 0$, then $\mathbf{d}_k$ is a descent direction (proved earlier)
- Line search with Wolfe conditions **guarantees global convergence**
- Near the solution: $\lambda_k \rightarrow 1$, so we get **local quadratic convergence** back
- Practical: **Most reliable** Newton implementation

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Levenberg-Marquardt Modification

Problem: The Hessian $\nabla^2 f(\mathbf{x}_k)$ may not be positive definite.

Consequence: Newton direction is not a descent direction (or undefined).

Solution: Regularize the Hessian

Levenberg-Marquardt Formula

$$M_k = \nabla^2 f(\mathbf{x}_k) + \mu_k I_n$$

where $\mu_k \geq 0$ is chosen so that $M_k \succ 0$.

The LM direction is:

$$\mathbf{d}_k^{\text{LM}} = -M_k^{-1} \nabla f(\mathbf{x}_k) = -[\nabla^2 f(\mathbf{x}_k) + \mu_k I_n]^{-1} \nabla f(\mathbf{x}_k)$$

How to choose μ_k :

- Compute minimum eigenvalue: $\lambda_{\min}(\nabla^2 f(\mathbf{x}_k))$
- Set $\mu_k = \max(0, \delta - \lambda_{\min})$ for small $\delta > 0$
- This ensures $M_k \succ 0$

LM: Descent Property

Theorem 6.1: Theorem: LM Direction is Descent

If $M_k = \nabla^2 f(\mathbf{x}_k) + \mu_k I \succ 0$ and $\nabla f(\mathbf{x}_k) \neq \mathbf{0}$, then

$$\mathbf{d}_k^{\text{LM}} = -M_k^{-1} \nabla f(\mathbf{x}_k)$$

is a descent direction.

Proof:

1. Compute directional derivative: $\theta'_k(0) = -\nabla f(\mathbf{x}_k)^T M_k^{-1} \nabla f(\mathbf{x}_k)$
2. Since $M_k \succ 0$, we have $M_k^{-1} \succ 0$
3. Therefore: $\nabla f(\mathbf{x}_k)^T M_k^{-1} \nabla f(\mathbf{x}_k) > 0$ for $\nabla f(\mathbf{x}_k) \neq \mathbf{0}$
4. Thus: $\theta'_k(0) < 0$ $\square$

With line search on the LM direction:

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \lambda_k M_k^{-1} \nabla f(\mathbf{x}_k)$$

Under suitable assumptions, LM enjoys global convergence!

LM: Interpolation Between SD and Newton

$$M_k = \nabla^2 f(\mathbf{x}_k) + \mu_k I_n$$

The LM method smoothly interpolates between two extremes:

- $\mu_k = 0$: Pure Newton method
 - Uses full Hessian information
 - Requires $\nabla^2 f(\mathbf{x}_k) \succ 0$
 - Quadratic convergence (if starting close)

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- $\mu_k \rightarrow \infty$: Approaches steepest descent
 - $M_k \approx \mu_k I$, so $M_k^{-1} \approx \mu_k^{-1} I$
 - Update: $\mathbf{x}_{k+1} \approx \mathbf{x}_k - \lambda_k \mu_k^{-1} \nabla f(\mathbf{x}_k)$
 - Linear convergence (but always guaranteed)

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 - Linear convergence (but always guaranteed)
- Adaptive μ_k :
 - Decrease μ_k when progress is good $\rightarrow$ shift toward Newton
 - Increase μ_k when progress is poor $\rightarrow$ shift toward steepest descent
 - Combines best of both worlds: global + local quadratic convergence

Exercise 2: Levenberg-Marquardt Direction

Computing the LM Direction For $f(x_1, x_2) = x_1^4 - 2x_1 + x_2^2$ at $\mathbf{x}_k = (1, 0)^T$:

1. Compute $\nabla f(\mathbf{x}_k)$ and $\nabla^2 f(\mathbf{x}_k)$. Is the Hessian positive definite?
2. Set $\mu_k = 2$ and compute the LM direction $\mathbf{d}_k^{\text{LM}}$
3. Compare with the Newton direction (if possible)

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Solution **(1)** $\nabla f(\mathbf{x}) = (4x_1^3 - 2, 2x_2)^T$, so $\nabla f(\mathbf{x}_k) = (2, 0)^T$

$$\nabla^2 f(\mathbf{x}) = \begin{pmatrix} 12x_1^2 & 0 \\ 0 & 2 \end{pmatrix}, \text{ so } \nabla^2 f(\mathbf{x}_k) = \begin{pmatrix} 12 & 0 \\ 0 & 2 \end{pmatrix} \succ 0 \text{ (PD)}$$

$$\textbf{(2)} \quad M_k = \begin{pmatrix} 14 & 0 \\ 0 & 4 \end{pmatrix}, \text{ so } M_k^{-1} = \begin{pmatrix} 1/14 & 0 \\ 0 & 1/4 \end{pmatrix}$$

$$\mathbf{d}_k^{\text{LM}} = - \begin{pmatrix} 1/7 \\ 0 \end{pmatrix}$$

(3) Since Hessian is PD, Newton direction =

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When Does Newton's Method Fail?

Challenge 1: Hessian Not Positive Definite

- Newton direction may not be a descent direction
- Can point uphill or be undefined
- **Solution:** Levenberg-Marquardt (add $\mu_k I$) or line search with modified Hessian

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- Without line search, Newton can diverge
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Challenge 3: Hessian Expensive to Compute

- For large n , computing and storing Hessian is prohibitive

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Conjugate Gradient for Coefficient

$$B_k \approx \nabla^2 f(\mathbf{x}_k).$$

Update B_k after each step:

$$\mathbf{x}_{k+1} = \mathbf{x}_k - B_k^{-1} \nabla f(\mathbf{x}_k)$$

==

Key Advantage:

- Cost: $O(n^2)$ per iteration (vs. $O(n^3)$ for Newton)
- Convergence: Superlinear (almost as fast as Newton)
- Storage: $O(n^2)$ for Hessian approximation

Popular Methods:

- BFGS (Broyden-Fletcher-Goldfarb-Shanno): maintains PD approximation, very robust

- L-BFGS: limited-memory variant for large-scale problems
- DFP (Davidon-Fletcher-Powell): historical (often inferior to BFGS)

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Key Takeaways: Newton's Method

Fundamental Properties

1. Minimizes local quadratic approximation of the objective
2. Quadratic convergence when starting close to the solution
3. One step to converge on quadratic functions
4. Requires both gradient and Hessian

Three Essential Modifications

1. Line Search: Ensures global convergence, preserves local quadratic rate
2. Levenberg-Marquardt: Handles indefinite Hessians, interpolates SD and Newton
3. Quasi-Newton (BFGS): Avoids $O(n^3)$ Hessian computation, maintains superlinear convergence

Next Topics

- Lecture 13: Quasi-Newton Methods (BFGS, DFP)
 - Approximate Hessian from gradient sequence
 - Superlinear convergence without Hessian computation
- Lectures 14-15: Conjugate Gradient Methods
 - Special structure for quadratic functions
 - Finite termination in n steps (in exact arithmetic)
 - Very efficient for large sparse problems
- Lectures 16+: Constrained Optimization
 - Projected gradient methods
 - Augmented Lagrangian methods
 - Interior point methods